

ACCOUNTING SOFTWARE BUILD PROJECT

1. Assignment Overview

In this project you will move beyond solving accounting problems by hand and instead build a small piece of software (a program, spreadsheet system, or web app) that performs the accounting work for you. Your software must take raw transaction data as input and automatically produce the ledgers, statements, and analyses that you have been calculating manually in class.

The assignment is organized into four modules that mirror the chapters we have covered. You may build one integrated system or four separate tools — either is acceptable as long as every requirement below is met.

2. Learning Objectives

- Demonstrate mastery of the accounting cycle (journalizing, posting, trial balance, adjusting entries, closing, and financial statements).
- Apply merchandising and inventory costing concepts (FIFO, LIFO, weighted average, COGS).
- Apply job orders to track product costs by job.
- Apply cost-volume-profit analysis to support business decision-making.
- Translate accounting logic into an automated, rule-based system — a core skill for modern accounting technology (ERP systems, Excel modeling, or accounting software like QuickBooks/Xero are all built on this same logic).

3. What You Are Building on: Module Breakdown

Module	Chapters Covered	What Software Must Do
1. Core Financial Accounting Engine	Ch. 1 Accounting in Action Ch. 2 The Recording Process Ch. 3 Adjusting the Accounts Ch. 4 Completing the Accounting Cycle	Users enter journal entries (with adjusting entries); the system automatically posts to the General Ledger, builds a Trial Balance (and Adjusted Trial Balance), and generates the Income Statement, Statement of Owner's Equity, and Balance Sheet.
2. Merchandising & Inventory Module	Ch. 5 Accounting for Merchandising Operations Ch. 6 Inventories	System records purchases, sales, purchase/sales returns and discounts; computes Cost of Goods Sold and ending inventory automatically under at least two costing methods (e.g., FIFO and Weighted Average) and produces a Merchandising Income Statement (multi-step).
3. Job Order Costing Module	Job Order Costing	System tracks direct materials, direct labor, and applied overhead for individual jobs; automatically generates a Job Cost Sheet per job and a Cost of Goods Manufactured

Module	Chapters Covered	What Software Must Do
		schedule, and flags over/under-applied overhead.
4. Cost-Volume-Profit (CVP) Module	Cost-Volume-Profit Analysis	Given selling price, variable cost, fixed costs, and unit/sales data, system automatically calculates contribution margin, break-even point (units and sales), margin of safety, and target-profit sales volume, ideally with a simple what-if / sensitivity option.

Module 1 — Core Financial Accounting Engine (Ch. 1–4)

Build the backbone of your system first — everything else conceptually sits on top of it.

1. Journal Entry Input: Create a form or input table where a user can enter a transaction (date, accounts debited/credited, amounts, description).
2. Automatic Posting: The system must automatically post each entry to the correct General Ledger accounts — do not let the user type ledger balances in manually.
3. Trial Balance: Automatically total debits and credits from the ledger and display a Trial Balance. It should visibly balance (or flag an error if it doesn't).
4. Adjusting Entries: Support at least the standard types — prepaid expenses, unearned revenue, accrued revenue, accrued expenses, and depreciation — and produce an Adjusted Trial Balance.
5. Financial Statements: Automatically generate the Income Statement, Statement of Owner's Equity, and Balance Sheet directly from the adjusted trial balance figures.
6. Closing Entries (recommended): Automatically close revenue, expense, and drawing accounts to Owner's Capital and produce a Post-Closing Trial Balance.

Module 2 — Merchandising & Inventory (Ch. 5–6)

Extend your system, or build a companion module, to handle a merchandising business.

1. Record purchase transactions (including purchase returns/allowances and purchase discounts) and sales transactions (including sales returns/allowances and sales discounts).
2. Automatically calculate Cost of Goods Sold and ending inventory value under at least two costing methods (e.g., FIFO and Weighted Average; LIFO optional).
3. Automatically generate a multi-step Income Statement showing Net Sales, COGS, Gross Profit, Operating Expenses, and Net Income.

Module 3 — Job Order Costing

Build a module that tracks the cost of individual custom jobs or batches.

1. Allow input of direct materials and direct labor charged to a specific job.
2. Apply manufacturing overhead automatically using a predetermined overhead rate that the user sets.
3. Automatically generate a Job Cost Sheet for each job (materials + labor + applied overhead = total job cost).
4. Automatically calculate over-applied or under-applied overhead when actual overhead is entered, and generate a simple Cost of Goods Manufactured schedule.

Module 4 — Cost-Volume-Profit (CVP) Analysis

Build a decision-support calculator, not just a static worksheet.

1. Accept selling price per unit, variable cost per unit, and total fixed costs as inputs.
2. Automatically calculate contribution margin (per unit and ratio), break-even point in units and sales dollars, margin of safety, and the sales volume needed to hit a target profit the user enters.
3. Recommended: let the user change any input and instantly see updated results (a live "what-if" calculator), rather than a one-time static calculation.

4. Technical Requirements

You may build your software using whichever tool you are most comfortable with, as long as the calculations are automated and traceable back to formulas or code — not manually typed final answers. Acceptable platforms include, for example:

- A spreadsheet system (Excel or Google Sheets) built entirely with linked formulas — no hard-typed results.
- A programming language / web app (Python, Java, JavaScript/HTML, etc.) with a simple interface.
- A no-code app builder, provided the logic is genuinely automated.

Whatever tool you choose, your software must be able to accept a brand-new set of transactions (that we will test with) and produce correct output — it cannot be pre-filled with only the example data you used to build it.

5. Deliverables & Submission

A. The Software Itself

- Submit the working file(s) or a link to your app/repository, exactly as instructed by your instructor (e.g., LMS upload, GitHub link, or shared Drive file).
- Include a short README or cover sheet (half a page) stating: the tool/language you used, how to open/run it, and any assumptions you made.

B. Demonstration Video (Maximum 3 Minutes)

Record a short screen-capture video (with your voice narrating) that proves your software works. In no more than 3 minutes you should:

1. Briefly introduce yourself and what tool/platform you built the system in (10–15 seconds).
2. Enter or point to a sample transaction and show it automatically flow through to the ledger/trial balance/statements (Module 1).
3. Quickly show the Merchandising/Inventory, Job Order Costing, and CVP modules producing correct automatic output.
4. Close by stating that all figures update automatically rather than being manually entered.

C. Upload Instructions

1. Name your video file exactly: LastName_FirstName_AccountingSoftwareDemo.mp4 (use your registered course name).
2. Upload the video to the shared Google Drive folder provided:
[\[https://drive.google.com/drive/folders/1sfkmOFxklmhsK1cRMYtHQj_zVIUVVd1y?usp=drive_link\]](https://drive.google.com/drive/folders/1sfkmOFxklmhsK1cRMYtHQj_zVIUVVd1y?usp=drive_link)
3. Do not change sharing permissions, rename, or delete other students' files in the shared folder.
4. After uploading, confirm your file appears in the folder **before the deadline (24/8/26 : 12am)** — upload failures after the deadline will not be accepted as an excuse for late submission.

6. Academic Integrity

You may use AI tools, templates, or tutorials to help you learn how to build formulas or code, but the accounting logic and final working system must be your own and you must be able to explain, in your video or if asked, why each part of your system calculates what it calculates. Copying another student's file structure or submitting a system you cannot explain is a violation of academic integrity.

7. Questions

If any requirement above is unclear, ask before the deadline — contact [khadija@iut-dhaka.edu and Mobile no: 01643570945].